

[Assignment - 2]

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section: 05

course : CSE330
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Initial

[Ans to the question no 1]

a

$$f(x) = \tan(x)$$

$$f'(x) = \sec^2 x$$

$$f''(x) = 2 \sec^2 x \tan x$$

$$f'''(x) = 2 \cdot \frac{d}{dx} [\sec^2 x \tan x]$$

$$= 2 \cdot \left[\frac{d}{dx} [\sec^2 x] \cdot \tan x + \sec^2 x \cdot \frac{d}{dx} [\tan x] \right]$$

$$= 2 (2 \sec x \cdot \sec x \cdot \tan x \cdot \tan x + \sec^2 x \cdot \sec^2 x)$$

$$= 2 (2 \sec^2 x \cdot \tan^2 x + \sec^4 x)$$

$$= 4 \sec^2 x \cdot \tan^2 x + 2 \sec^4 x$$

$$= \cancel{2 \sec^2 x} (4 \tan^2 x + \sec^2 x)$$

Evaluating at, $x=0$,

$$f(0) = \tan(0) = 0$$

$$f'(0) = \sec^2(0) = 1$$

$$f''(0) = 2 \sec^2(0) \tan(0) = 0$$

$$f'''(0) = 2 \sec^4(0) + 4 \sec^2(0) \cdot \tan^2(0) = 2(1) + 0 = 2$$

The Taylor polynomial is :

$$\begin{aligned} P_3(x) &= f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 + \frac{f'''(0)}{3!}(x-0)^3 \\ &= 0 + x + 0 + \frac{2x^3}{3!} \end{aligned}$$

∴ Taylor coefficients:

$$a_0 = 0$$

$$a_1 = 1$$

$$a_2 = 0$$

$$a_3 = \frac{1}{3}$$

(Ans)

b

$$f(x) = \tan(x)$$

$$\therefore f\left(\frac{\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) = 1$$

$$\begin{aligned} P_3(x) &= x + \frac{2x^3}{6} \\ &= x + \frac{x^3}{3} \end{aligned}$$

$$\therefore P_3\left(\frac{\pi}{4}\right) = \frac{\pi}{4} + \frac{\left(\frac{\pi}{4}\right)^3}{3} = \underline{\underline{0.7}}$$

$$= 0.7854 + 0.1615 = 0.785 + 0.161$$

$$= 0.947$$

$$\text{Percentage relative error} = \frac{\left| f\left(\frac{\pi}{4}\right) - P_3\left(\frac{\pi}{4}\right) \right|}{f\left(\frac{\pi}{4}\right)} \times 100$$

$$= \frac{|1 - 0.947|}{1} \times 100$$

$$= 5.3 \%$$

(Ans)

C

$$f(x) = \tan x$$

$$p_3(x) = x + \frac{x^3}{3}$$

Lagrange remainder form is:

$$R_3(x) = \left| \frac{f^{(4)}(c)}{4!} (x)^4 \right|$$

$$\begin{aligned} f^{(4)} &= \frac{d}{dx} (4 \sec^2 x \cdot \tan^2 x + 2 \sec^4 x) \\ &= 4 \cdot \frac{d}{dx} [\sec^2 x \cdot \tan^2 x] + 2 \cdot \frac{d}{dx} [\sec^4 x] \\ &= 4 \left(\frac{d}{dx} [\sec^2 x] \cdot \tan^2 x + \sec^2 x \cdot \frac{d}{dx} [\tan^2 x] \right) \\ &\quad + 2 \cdot 4 \sec^3 x \cdot \frac{d}{dx} (\sec x) \\ &= 4 (2 \sec x \cdot \sec x \cdot \tan x \cdot \tan^2 x + \sec^2 x \cdot \sec^2 x \cdot 2 \tan x) + 8 \sec^4 x \tan x \\ &= 4 (2 \sec^2 x \cdot \tan^3 x) + 8 \sec^4 x \tan x \\ &\quad + 8 \sec^4 x \tan x \\ &= 16 \sec^4 x \tan x + 8 \sec^2 x \cdot \tan^3 x \end{aligned}$$

To find the maximum value of $g^4(\frac{\pi}{4})$ on $[0, \frac{\pi}{4}]$, we note that both $\sec x$ and $\tan x$ are increasing on this interval.

$$\text{At, } \frac{\pi}{4} = \frac{\pi}{4},$$

$$g^4\left(\frac{\pi}{4}\right) = 16\left(\sec\frac{\pi}{4}\right)^4 \cdot \tan\left(\frac{\pi}{4}\right) + 8\left(\sec\frac{\pi}{4}\right)^2 \cdot \left(\tan\left(\frac{\pi}{4}\right)\right)^3$$

$$= 16(\sqrt{2})^4 \cdot 1 + 8(\sqrt{2})^2 \cdot 1$$

$$= 80$$

$$\therefore R_3\left(\frac{\pi}{4}\right) = \left| \frac{80}{4!} \left(\frac{\pi}{4}\right)^4 \right|$$

$$= 1.27$$

(Ans)

[Ans to the question no 2]

9

$$f(x) = e^x - e^{-x}$$

$$\text{and, } \left. \begin{array}{l} x_0 = -1 \\ x_1 = 0 \\ x_2 = 1 \end{array} \right\} \text{Nodes}$$

$$f(-1) = e^{-1} - e^1 = -2.350$$

$$f(0) = e^0 - e^{-0} = 0$$

$$f(1) = e^1 - e^{-1} = 2.350$$

$$P(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

$$P_2(x) = a_0 x_0^0 + a_1 x_0^1 + a_2 x_0^2 = -2.350$$

$$= a_0 + a_1(-1) + a_2(-1)^2$$

$$\therefore P_2(x) = a_0 x_1^0 + a_1 x_1^1 + a_2 x_1^2 = 0$$

$$P_2(x) = a_0 x_2^0 + a_1 x_2^1 + a_2 x_2^2 = 2.350$$

$$\Rightarrow \begin{pmatrix} x_0^0 & x_0^1 & x_0^2 \\ x_1^0 & x_1^1 & x_1^2 \\ x_2^0 & x_2^1 & x_2^2 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -2.35 \\ 0 \\ 2.35 \end{pmatrix}$$

(V) Vandermonde's matrix A F

matrix b : $b = [-2.350, 0, 2.350]^T$

vander monde matrix, $V = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

b

$$V = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} \det(V) &= 1(0 - 0) - 1(0 - 1) + 1(1 - 0) \\ &= 0 + 1 + 1 \\ &= 2 \end{aligned}$$

(Ans)

c

$$f(x) = e^x - e^{-x}$$

$$f'(x) = e^x + e^{-x}$$

$$f''(x) = e^x - e^{-x}$$

$$f(0) = e^0 - e^{-0} = 0$$

$$f'(0) = e^0 + e^{-0} = 2$$

$$f''(0) = 0$$

$$\therefore f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2$$

$$= 0 + 2x + 0$$

$$= 2x$$

Taylor coefficients are,

$$\therefore a_0 = 0$$

$$a_1 = 2$$

$$a_2 = 0$$

$$\therefore P(x) = 0x^0 + 2x^1 + 0x^2$$

$$= 2x$$

$$f(x) = e^x - e^{-x}$$

$$f'(x) = e^x + e^{-x}$$

$$f''(x) = e^x - e^{-x}$$

$$f'''(x) = e^x + e^{-x}$$

2

$$\begin{aligned}
 f'''(2.1) &= e^{2.1} + e^{-2.1} \\
 &= 8.166 + 0.122 \\
 &= 8.288 \\
 &= 8.29
 \end{aligned}$$

$$\begin{aligned}
 |S(x) - P_n(x)| &\leq \frac{f^{n+1}(\xi)}{(n+1)!} (x-x_0)(x-x_1)(x-x_2) \\
 &\leq \frac{e^{2.1} + e^{-2.1}}{3!} (x+1)x(x-1) \\
 &\leq \frac{8.29}{3!} \times 7.16
 \end{aligned}$$

$$\begin{aligned}
 \omega(x) &= (x+1)(x-1)x \\
 &= (x^2-1)x
 \end{aligned}
 \quad \therefore \leq 9.89$$

$$\omega(x) = x^3 - x$$

$$\omega'(x) = 3x^2 - 1$$

Now,

$$3x^2 - 1 = 0$$

$$\Rightarrow x = \frac{\sqrt{3}}{3} \text{ or } -\frac{\sqrt{3}}{3}$$

$x \rightarrow \omega(x)$	$x \rightarrow \omega(x)$
$\frac{\sqrt{3}}{3} \rightarrow -0.385$	$2.1 \rightarrow 7.16$
$-\frac{\sqrt{3}}{3} \rightarrow 0.385$	$-2.1 \rightarrow -7.16$

[Ans to the question no 3]

9

$$s(x) = e^x + e^{-x}$$

$$s(-1) = e^{-1} + e^1$$

$$s(0) = 2$$

$$s(1) = e^1 + e^{-1}$$

$$x_0 = -1$$

$$x_1 = 0$$

$$x_2 = 1$$

$$p_2(x) = s(x_0) l_0(x) + s(x_1) l_1(x) + s(x_2) l_2(x)$$

Here,

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2}$$

$$= \frac{x - 0}{-1 - 0} \cdot \frac{x - 1}{-1 - 1}$$

$$= \frac{x}{-1} \cdot \frac{x - 1}{-2} = \frac{x(x - 1)}{2}$$

$$l_1(x) = \frac{x - x_2}{x_1 - x_2} \cdot \frac{x - x_0}{x_1 - x_0}$$

$$= \frac{x - 1}{0 - 1} \cdot \frac{x + 1}{0 + 1}$$

$$= \frac{x - 1}{-1} \cdot \frac{x + 1}{1}$$

$$= -(x^2 - 1)$$

$$\begin{aligned}
 l_2(x) &= \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} \\
 &= \frac{x+1}{1+1} \cdot \frac{x-0}{1-0} \\
 &= \frac{x(x+1)}{2}
 \end{aligned}$$

b

$$\begin{aligned}
 P_2(x) &= f(-1) l_0(x) + f(0) l_1(x) + f(1) l_2(x) \\
 &= 3.09 \times \frac{x(x-1)}{2} + 2(-x^2-1) + 3.09 \times \frac{x(x+1)}{2} \\
 &= 1.55x^2 - 1.55x + 2(-x^2+1) + 1.55x \\
 &\quad + 1.55x^2 \\
 &= 3.1x^2 - 2x^2 + 2 \\
 &= 1.1x^2 + 2
 \end{aligned}$$

$$\begin{aligned}
 P_2(6) &= 1.1 \times (6)^2 + 2 \\
 &= 39.6 + 2 \\
 &= 41.6
 \end{aligned}$$

$$\begin{aligned}
 f(6) &= e^6 + e^{-6} \\
 &= 403
 \end{aligned}$$

9

$$S(1.5) = e^{1.5} + e^{-1.5}$$
$$= 4.708 \approx 4.70$$

$$P(1.5) = 1.18(1.5)^2 + 2$$
$$= 4.48 \approx 4.44$$

$$\therefore \text{Relative error} = \frac{|S(1.5) - P(1.5)|}{S(1.5)} \times 100$$
$$= \frac{(4.708 - 4.48)}{4.708} \times 100$$
$$= 5.55\% \quad \text{--- } 5.73\%$$
$$= 4.68\%$$

(Ans)

[Ans to the question no 4]

9

$$S(x) = e^x - e^{-x}$$

$x_0 = -2$	$S(-2) = -7.254 \approx 7.25$
$x_1 = 0$	$S(0) = 0$
$x_2 = 2$	$S(2) = 7.254 \approx 7.25$

Divided-Difference method :

$$x_0 = -2 \quad f[x_0] = -7.25$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{7.25}{2} = 3.63$$

$$x_1 = 0 \quad f[x_1] = 0$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{0}{4} = 0$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{7.25}{2} = 3.63$$

$$x_2 = 2 \quad f[x_2] = 7.25$$

$$p_2(x) = a_0 n_0(x) + a_1 n_1(x) + a_2 n_2(x)$$

$$= f[x_0] + f[x_0, x_1](x - x_0)$$

$$+ f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$= -7.25 + 3.63(x + 2) + 0$$

So,

$$a_0 = -7.25$$

$$a_1 = 3.63$$

$$a_2 = 0$$

b

$$\begin{aligned}P_2(x) &= a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) \\&= -7.25 + 3.63(x+2) \\&= -7.25 + 3.63x + 7.26 \\&= 0.01 + 3.63x\end{aligned}$$

$$P_2(6) = 3.63(6) + 0.01 = 21.8$$

(Ans)

$$\begin{aligned}f(6) &= e^6 - e^{-6} \\&= 403,\end{aligned}$$

c

$$\begin{aligned}f(1.5) &= e^{1.5} - e^{-1.5} \\&= 4.26\end{aligned}$$

$$P(1.5) = 3.63(1.5) + 0.01 = 5.46$$

$$\begin{aligned}\therefore \text{Relative Error} &= \frac{|f(1.5) - P(1.5)|}{f(1.5)} \times 100 \\&= \frac{|4.26 - 5.46|}{4.26} \times 100 \\&= 28.2\%.\end{aligned}$$

(Ans)